AAE 451 - Aircraft Conceptual Design

3D-Printed Aircraft Design Course Notes

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0.1 Lecture Notes

0.1.1 Initial Vehicle Sizing by Constraint Analysis

Purpose of the Constraint Diagram

The correct sizing of an airplane depends on numerous important variables. A clearly stated mission plays a paramount role in this respect and allows the sizing to be accomplished using mathematical tools. Constraint analysis is used to assess the relative significance of selected aircraft performance parameters on the design with respect to a set of requirements.

Specifically, it allows one to understand which combinations of wing loading (W/S) and power loading (P/W) permit the aircraft to simultaneously meet dissimilar performance requirements, such as rate of climb, take-off distance, and others.

Overview of Constraint Analysis

Given mission requirements, we perform a constraint analysis using a constraint diagram. From this, we select design parameter values such as:

- Stall Speed (V_s)
- Cruise Speed (V_{cr})
- Climb rate or climb angle
- Take-off run
- Maneuver loads

Design Space: The design space is that region within which all the constraints are met. For each quantitative requirement of the mission, we find the relationship between the wing loading and the power loading that exactly satisfies that requirement.

The general form of the Constraint Equation is:

$$\left(\frac{P}{W}\right)_{Sea-Level} = f\left(\frac{W}{S}\right) \tag{1}$$

For each requirement, we find the region where the vehicle violates that requirement and indicate the forbidden region with a hash mark. The feasible region (where we want to design our aircraft) is the intersection of all allowed regions. We typically prefer smaller engines (lower P/W for cost/weight) and smaller wings (higher W/S), so the optimal point often lies at a corner of the feasible space.

1st Constraint: Sizing to Stall Speed

To satisfy the stall speed requirement, the aircraft must be able to generate enough lift to balance its weight at stall velocity V_s.

$$L = W = \frac{1}{2}\rho V_s^2 S_{ref} C_{L_{max}} \tag{2}$$

Rearranging for Wing Loading (W/S):

$$\frac{W}{S} = \frac{1}{2}\rho V_s^2 C_{L_{max}} \tag{3}$$

Where:

- W = weight
- S = wing area (S_{ref})
- ρ = air density
- V_s = stall velocity
- $C_{L_{max}}$ = maximum lift coefficient

This is a vertical line on the constraint diagram. Any wing loading greater than this value will violate the stall speed requirement.

2nd Constraint: Sizing to Cruise Speed

Assuming the aircraft is in level flight during cruise, the power required (P_{req}) must be less than or equal to the power available from the engine ($P_{avail} \cdot \eta_p$).

$$P_{req} = D \cdot V_{cr} = \frac{1}{2}\rho V_{cr}^3 S C_D \tag{4}$$

Where:

- η_p is the propeller efficiency
- C_D is the drag coefficient

Cruise speeds for propeller-driven airplanes are typically calculated at 75 to 80 percent maximum power. Often, induced drag is small compared to profile drag, giving:

$$C_D = C_{D_0} + 0.1C_{D_0} = 1.1C_{D_0} \tag{5}$$

If cruise is at sea-level, the constraint equation is:

$$\frac{P}{W} \geq \frac{1.1\rho V_{cr}^3 C_{D_0}}{2\eta_p\left(\frac{W}{S}\right)} \tag{6}$$

This equation creates a curve on the constraint diagram where power loading must be greater than or equal to the required power loading for a given wing loading.

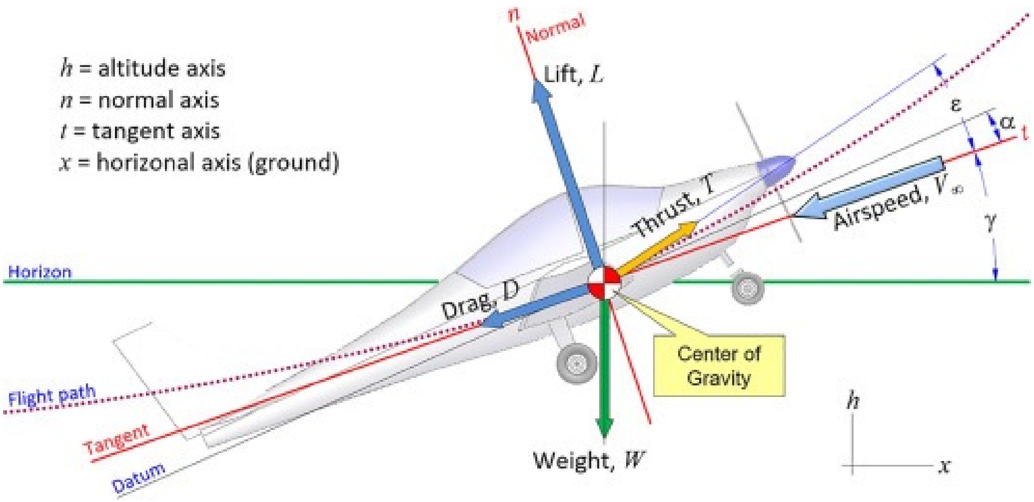

Figure 1: Constraint diagram showing the stall speed and cruise speed constraints with the feasible design region.

3rd Constraint: Sizing to Climb Requirement

For an aircraft in a climb:

- Thrust direction: $T\cos(\alpha + \epsilon_T) - D = W\sin(\gamma)$
- Lift direction: $L + T\sin(\alpha + \epsilon_T) = W\cos(\gamma)$

Assuming $\alpha + \epsilon_T$ is small and $T\sin(\alpha + \epsilon_T)/W$ is small, we have:

$$L \approx W \tag{7}$$

$$D = \frac{W}{L/D} \tag{8}$$

Climbing at $L/D = 0.866(L/D)_{max}$, the thrust required is:

$$T = \frac{W}{0.866(L/D)_{max}} + W\sin(\gamma) \tag{9}$$

Table 1: Variables for the Maneuver Constraint

Variable	Description
$q = \frac{1}{2}\rho V^2$	Dynamic pressure
n	Load factor
AR	Aspect ratio
e	Oswald efficiency factor

Accounting for propeller efficiency $\eta_p = \frac{T \cdot V_{climb}}{P}$:

$$\frac{P}{W} \geq \frac{V_{climb}}{\eta_p}\left(\frac{1}{0.866(L/D)_{max}} + \sin(\gamma)\right) \tag{10}$$

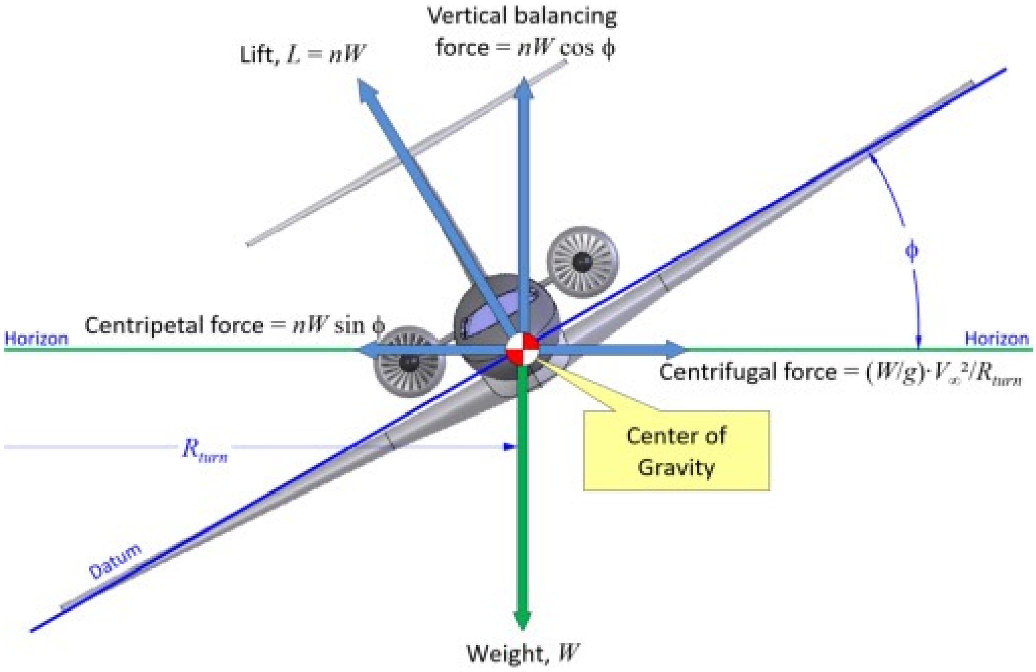

Figure 2: Constraint diagram including the climb constraint, illustrating how the design space narrows with each additional requirement.

4th Constraint: Sizing to Maneuver Requirement

Consider an airplane in a level constant velocity turn. The load factor is $n = 1/\cos(\phi)$, where ϕ is the bank angle.

The required thrust to maintain load factor n while banking at constant airspeed and altitude is:

$$T = qSC_{D_0} + \frac{n^2W^2}{qS\pi eAR} \tag{11}$$

Dividing by weight and multiplying by velocity to get power:

$$\frac{P}{W} \geq \frac{V}{\eta_p}\left(\frac{qC_{D_0}}{W/S} + \frac{n^2(W/S)}{q\pi eAR}\right) \tag{12}$$

5th Constraint: Take-Off

During the take-off run (S_{to}), the aircraft accelerates from $V = 0$ to $V = V_{to}$ (typically $1.1V_s$ to $1.3V_s$). The general form for the take-off constraint will depend on runway friction (μ) and aerodynamic forces during the ground roll.

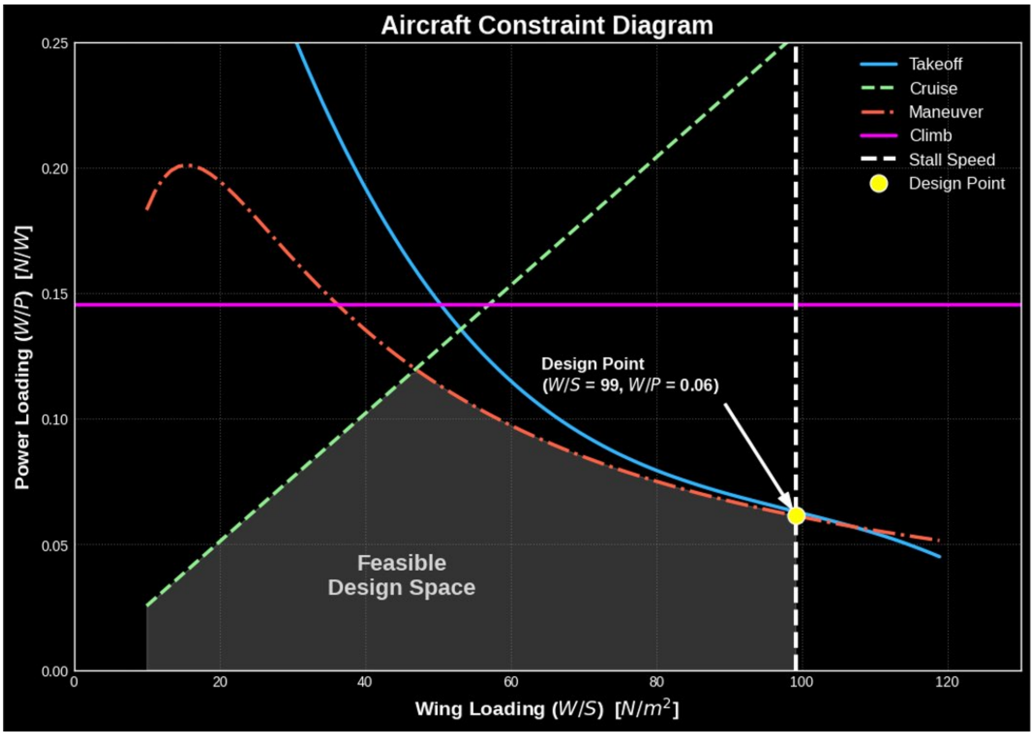

Figure 3: Complete constraint diagram with all four constraints (stall, cruise, climb, maneuver) showing the final feasible design space.

Table 2: Variables for the Take-Off Constraint

Variable	Description
S_{to}	Maximum take-off distance
C_{D_g}, C_{L_g}	Drag and lift coefficients during ground roll
μ	Runway friction coefficient
g	Gravitational acceleration

Based on Sadraey's derivation (Chapter 4), the power loading constraint is:

$$\frac{W}{P} \leq \frac{\eta_P}{V_{to}} \left[\frac{1 - \exp\left(\frac{0.6 g \rho C_{D_g} S_{to}}{W/S}\right)}{\mu - \left(\mu + \frac{C_{D_g}}{C_{L_g}}\right) \exp\left(\frac{0.6 g \rho C_{D_g} S_{to}}{W/S}\right)} \right] \tag{13}$$

The Master Constraint Diagram

By plotting all the above constraints (P/W vs. W/S), we identify the **Design Space**. The preferred design point is usually the one that minimizes the engine size (lowest P/W) and minimizes wing size (highest W/S) while staying within the feasible region.

> **Note:** See the Jupyter Notebook `DBF_Constraint_Diagram.ipynb` for the Python implementation and visualization of these constraints.

Reading & References

- **Gundlach:** Ch. 3, 9.4
- **Roskam (Vol. I):** Ch. 3
- **Gudmundsson:** Ch. 3
- **Sadraey:** Ch. 4

Table 3: Typical Mission Profile Phases

Phase	Description
1	Warmup and takeoff
2	Climb
3	Cruise (Level flight)
4	Turning flight (if applicable)
5	Descent
6	Land and taxi

Table 4: Weight Component Definitions

Symbol	Description
W_e	Empty weight (structure, landing gear, motor, etc.)
W_B	Battery weight
W_P	Payload weight

0.1.2 Initial Vehicle Weight Estimation

Our Approach

In this chapter, we focus on the initial weight estimation of electric-powered aircraft. The primary question is: **How much energy is needed to complete a design mission?**

To estimate this energy, our approach is as follows:

1. Consider how much energy is used in each mission phase.
2. Estimate the battery weight required to store that much energy.
3. Use historical data to determine the empty weight of the aircraft required to carry that battery plus any payload.

Weight Breakdown

The total aircraft weight (W) is broken down into three main components:

$$W = W_e + W_B + W_P \tag{14}$$

Empty Weight Prediction

Empty weight (W_e) consists of anything that cannot be removed from the aircraft without a tool. Early in the design process, when very little detail about the aircraft is known, we must use predictors or **Weight Estimating Relationships (WER)** based on historical data of similar classes of aircraft.

This historical data is usually represented as the empty weight fraction (W_e/W) plotted against the total weight (W), which typically yields a curve fit that we can use for estimation.

For a battery-powered, fixed-wing remotely controlled airplane, specific historical relations for $W_e/W = f(W)$ can be established.

Power Flow

To understand energy consumption, we must first understand the flow of power from the battery to the aircraft.

Therefore, the total power required from the battery is related to the required thrust power by:

Table 5: Power Flow Chain

Stage	Power	Efficiency	Range
Battery Output	P_B	—	—
Motor Output	$P_m = \eta_m \cdot P_B$	η_m	0.7–0.9
Propeller Output	$P_r = \eta_p \cdot P_m$	η_p	0.5–0.9

$$P_r = \eta_p \cdot \eta_m \cdot P_B \quad \Rightarrow \quad P_B = \frac{P_r}{\eta_p \cdot \eta_m} \tag{15}$$

Energy and Mission Phases

Energy (E) is Power (P) multiplied by time (t):

$$E_{battery} = P_B \cdot t = \frac{P_r \cdot t}{\eta_p \eta_m} \tag{16}$$

We divide the mission into various flight phases and determine the energy required for each segment:

$$E_{total} = E_{to} + E_{wu} + E_{cl} + E_{lf} + E_{tf} \tag{17}$$

The total energy required by the battery (E_B) determines the battery weight (W_B) through the **energy density** of the battery (e_B, in J/kg; e.g., Lithium Polymer $\approx 5.27 \times 10^5$ J/kg):

$$W_B = \frac{E_{total}}{e_B} \tag{18}$$

Since the individual energy components are functions of the total aircraft weight (which is yet to be determined), we normalize by the unknown aircraft weight W. The problem boils down to finding the **battery weight fractions** ($W_{B,i}/W$) for each mission phase:

$$\frac{W_B}{W} = \frac{W_{B,to}}{W} + \frac{W_{B,wu}}{W} + \frac{W_{B,cl}}{W} + \frac{W_{B,lf}}{W} + \frac{W_{B,tf}}{W} \tag{19}$$

Energy Consumption in Level Flight

In level flight, Lift equals Weight ($L = W$) and Thrust equals Drag ($T = D$). The thrust power required is:

$$P_r = T \cdot V = D \cdot V = \frac{W}{L/D} \cdot V \tag{20}$$

The energy required for a level flight time t_{lf} is $E_{lf} = P_r \cdot t_{lf}$. Factoring in the efficiencies and energy density, the battery weight fraction for level flight is:

$$\boxed{\frac{W_{B,lf}}{W} = \frac{V \cdot t_{lf}}{\eta_p \eta_m (L/D)\, e_B}} \tag{21}$$

Energy Consumption in Turning Flight

In a turning flight with bank angle ϕ and turning radius R, the load factor $n = 1/\cos\phi$. The lift increases, and consequently, drag increases by a factor of n.

$$\boxed{\frac{W_{B,tf}}{W} = \frac{V \cdot t_{tf} \cdot n}{\eta_p \eta_m (L/D)\, e_B}} \tag{22}$$

Table 6: Turning Flight Variables

Variable	Description
R	Turning radius (m)
ϕ	Bank angle (rad)
$n = 1/\cos\phi$	Load factor
$g = 9.81$ m/s^2	Gravitational acceleration

Table 7: Solution Procedure Steps

Step	Action
1	For a given payload W_P, calculate the total battery weight fraction W_B/W from Eq. (19)
2	Compute battery weight: $W_B = (W_B/W) \times W$
3	Rearrange the weight equation: $W - W_e = W_B + W_P$
4	Plot $(W_B + W_P)$ and $(W - W_e)$ vs. total weight W using the historical W_e/W curve
5	The intersection gives the **Initial Weight Estimate**

Energy Consumption in Climbing Flight

In a climb to altitude h at climb angle γ, the sum of forces yields:

$$L = W\cos\gamma, \qquad T = D + W\sin\gamma \tag{23}$$

Using $D = L/(L/D) = W\cos\gamma/(L/D)$, the power required is:

$$P_r = T \cdot V_{cl} = V_{cl} \cdot W\left(\frac{\cos\gamma}{L/D} + \sin\gamma\right) \tag{24}$$

For a time to climb $t_{cl} = h/(V_{cl}\sin\gamma)$, the battery weight fraction is:

$$\boxed{\frac{W_{B,cl}}{W} = \frac{V_{cl} \cdot t_{cl}}{\eta_p\eta_m\, e_B}\left(\frac{\cos\gamma}{L/D} + \sin\gamma\right)} \tag{25}$$

Energy Consumption in Warmup and Take-off

During take-off for a time t_{to}, the battery output power is $P_B = P_m/\eta_m$. Normalizing by the airplane weight to get the inverse of the power loading (P_m/W), the battery weight fraction for take-off is:

$$\boxed{\frac{W_{B,to}}{W} = \frac{t_{to}}{\eta_m\,(W/P)\;e_B}} \tag{26}$$

For warmup, a fraction N of the take-off energy is typically assumed:

$$\frac{W_{B,wu}}{W} = N \cdot \frac{W_{B,to}}{W} \tag{27}$$

Solution Procedure Summary

Suggested Reading

- **Gundlach:** Ch. 3
- **Raymer:** Ch. 3 and Ch. 20.11

Table 8: Wing Design Parameters to Determine

Parameter	Symbol	Description
Reference area	S_{ref}	Wing planform area
Aspect ratio	$AR = b^2/S$	Span-to-chord ratio
Taper ratio	$\lambda = c_{tip}/c_{root}$	Tip-to-root chord ratio
Sweep	Λ	Leading or quarter-chord sweep angle
Dihedral	Γ	Upward angle of wing panels
Twist (washout)	ϵ_t	Spanwise twist angle
Airfoil(s)	—	Cross-section profile
Incidence	i_w	Wing setting angle relative to fuselage
Flap & aileron	—	High-lift and lateral control devices
MAC	$\bar{c}$	Mean Aerodynamic Chord

Table 9: High Wing vs. Low Wing Trade-offs

High Wing	Low Wing
Increases $C_{L_{max}}$ (ground effect)	Better take-off performance
Aircraft is laterally more stable	Less frontal area
Produces slightly more lift	Less induced drag
Usually lower stall speed	More lateral control
Drag produces nose-down pitching moment	Less stable ($C_{l_\beta} > 0$)
Less downwash on the tail	More downwash effect on the tail
Less lateral control	Less ground effect for take-off/landing

0.1.3 Wing and Tail Design

Wing Design – Main Definitions

Lift comes with a price: **drag** and a **nose-down pitching moment**. Wing design decisions should therefore reflect the performance requirements, including stall speed, maximum speed, take-off run, range, and endurance.

Wing Arrangement

Number of Wings

- **Monoplane (single wing):** the practical and modern solution, enabled by advanced materials.
- **Biplane:** used when span is limited; shorter span delivers higher roll control, but at higher weight.

Wing Location

Airfoil Selection

The airfoil is the fundamental building block of the wing. Selection criteria must balance competing objectives.

Wing Incidence The wing incidence angle i_w should be set such that the fuselage generates **minimum drag during cruise**, i.e., the fuselage angle of attack is approximately $0 deg$ during the cruise condition.

Table 10: Airfoil Selection Criteria

Objective	Design Driver
Maximum lift ($C_{l_{max}}$)	Take-off and landing performance, stall speed
Minimum drag ($C_{d_{min}}$)	Range, endurance, max speed
Low pitching moment ($C_{m_{ac}}$)	Reduced trim drag, smaller tail
Gentle stall characteristics	Safety and recoverability

Representative Aerodynamic Coefficients of Selected Airfoils (Low Re ≈ 200k)

	Airfoil	CLo	CMo	CLα (per rad)	CL,max (approx)	α_stall (deg)	Max L/D (approx)	α for Max L/D (deg)	CD,min (approx)
0	Selig S1223	1.750	-0.210	6.400	2.200	15	45	8.000	0.019
1	Eppler E423	1.420	-0.200	6.300	1.850	14	60	6.000	0.012
2	Selig S1210	1.500	-0.160	6.500	2.000	16	40	9.000	0.018
3	Eppler 205	0.480	-0.100	6.100	1.450	13	80	3.000	0.009
4	NACA 2412	0.240	-0.045	6.000	1.550	15	65	4.000	0.008
5	SD7062	0.780	-0.120	6.200	1.500	14	60	6.000	0.010
6	Clark Y	0.360	-0.050	5.700	1.520	16	70	4.500	0.009
7	MH 32	0.220	-0.057	6.200	1.350	12	90	2.500	0.007

Figure 4: Representative aerodynamic coefficients of selected airfoils at low Reynolds number ($Re \approx$ 200,000), comparing lift-curve slope, maximum C_L, stall angle, maximum L/D, and minimum drag.

Considerations about Aspect Ratio

The aspect ratio AR has a fundamental impact on the aerodynamic performance of the wing.

The **3D wing lift-curve slope** is related to the 2D airfoil lift-curve slope C_{l_α} by:

$$C_{L_\alpha} = \frac{C_{l_\alpha}}{1 + \frac{C_{l_\alpha}}{\pi \cdot AR}} \tag{28}$$

The **induced drag factor** is:

$$K = \frac{1}{\pi \cdot e \cdot AR} \tag{29}$$

And the **maximum lift-to-drag ratio** is:

$$\left(\frac{L}{D}\right)_{max} = \frac{1}{2\sqrt{K \cdot C_{D_0}}} \tag{30}$$

Where C_{D_0} is the zero-lift drag coefficient, and e is the Oswald efficiency factor.

> **Design rule:** The horizontal tail should have an AR lower than the wing to ensure the tail stalls after the wing (no twist required on the tail).

Considerations about Taper Ratio

The taper ratio $\lambda = c_{tip}/c_{root}$ influences the spanwise lift distribution, structural weight, and manufacturing complexity.

Table 11: Effects of Aspect Ratio

Low AR	High AR
Increases stall angle α_{stall}	Higher L/D
Reduces C_{L_α}	Higher C_{L_α}
Lower structural weight	Higher structural weight
Better roll rate	Longer span, higher bending loads

Table 12: Effects of Taper Ratio

Effect	Description
$\lambda \approx 0.4 - -0.5$	Approximates elliptical lift distribution
$\lambda < 0.4$	Reduces weight, but tip stall risk increases
$\lambda = 1.0$	Rectangular wing — easy to manufacture
Decreases I_{xx}	Improves lateral control (roll rate)

Table 13: Tail Design Parameters

Horizontal Tail	Vertical Tail
S_t — planform area	S_v — planform area
b_t — span	b_v — height
l_t — tail arm	l_v — tail arm
Λ_t — sweep	Λ_v — sweep
Airfoil	Airfoil
AR_t	AR_v
λ_t — taper ratio	λ_v — taper ratio
i_t — incidence	—

$$\bar{c} = \frac{2}{3} c_{root} \frac{1 + \lambda + \lambda^2}{1 + \lambda} \tag{31}$$

Considerations about Twist (Washout)

Wing twist (washout) reduces the angle of attack at the tip relative to the root. This serves two purposes:

1. **Prevents tip stall before root stall**, which is critical for maintaining aileron effectiveness and safe stall recovery.
2. **Helps approximate an elliptical spanwise lift distribution**, reducing induced drag.

A typical washout angle for general aviation is $\epsilon_t \approx 2deg - -5deg$.

Tail Design

The horizontal and vertical tails provide pitch and yaw stability and control. The key parameters to determine are listed below.

Pitching Moment and Longitudinal Stability

The total pitching moment about the center of gravity (M_{cg}) is the sum of contributions from the wing, fuselage, propulsion, and tail:

$$C_{M_{cg}} = C_{M_{ac,w}} + C_{M_{fus}} + C_{M_{prop}} + C_{M_{tail}} \tag{32}$$

Expanding:

$$M_{cg} = M_{ac,w} + (x_{cg} - x_{ac})L_w + M_{ac,t} - l_t L_t + M_F + M_P \tag{33}$$

The **pitching moment coefficient** about the c.g. is:

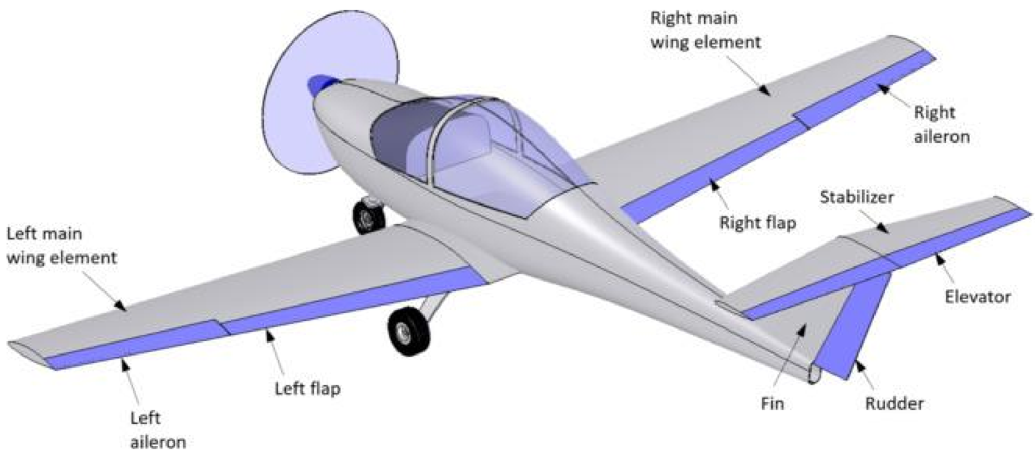

Figure 5: Aircraft control surfaces: ailerons, flaps, elevator, rudder, fin, and stabilizer (Source: Gudmundsson).

$$C_{M_{cg}} = C_{M_{ac,w}} + \left(C_{L_W} + C_{L_t}\frac{S_t}{S_w}\right)\frac{(\bar{x}_{cg} - \bar{x}_{ac,w})}{\bar{c}} - C_{L_t}V_H + C_{M_{ac,t}}V_H\frac{\bar{c}_t}{l_t} \tag{34}$$

Where the **horizontal tail volume coefficient** is defined as:

$$\boxed{V_H = \frac{S_t \cdot l_t}{S_w \cdot \bar{c}}} \tag{35}$$

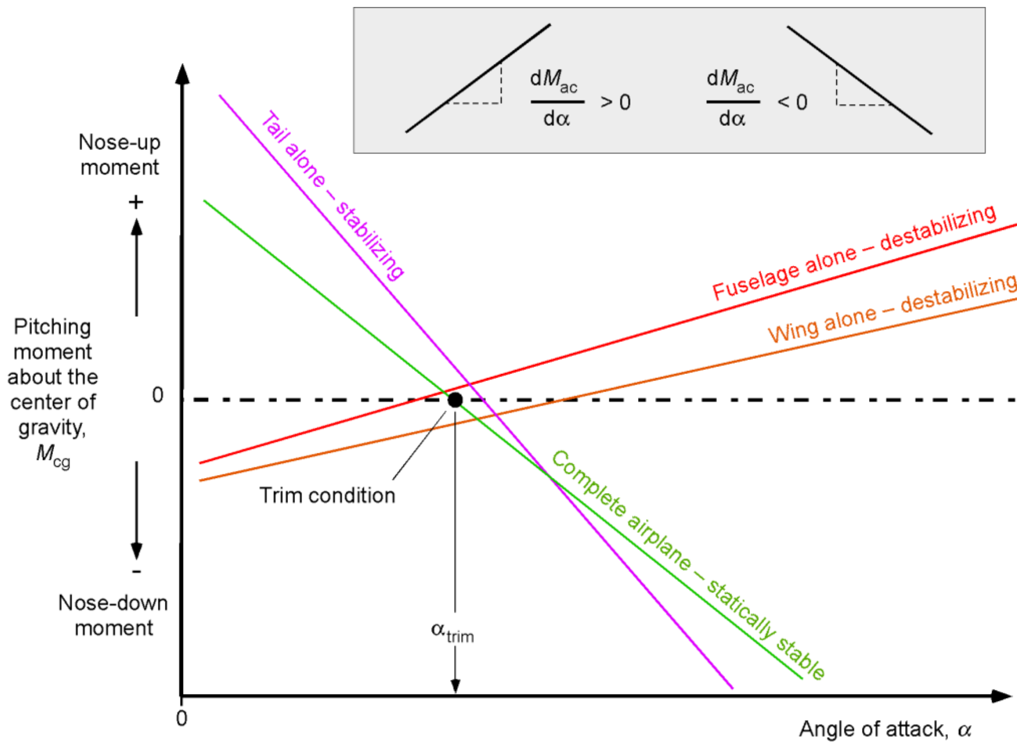

Figure 6: Pitching moment contributions from each aircraft component. The tail provides a stabilizing (positive slope) contribution, while the fuselage and wing alone are destabilizing. The complete aircraft curve must have a negative slope ($dM_{ac}/d\alpha < 0$) for static stability. The trim condition occurs where $M_{cg} = 0$.

Table 14: Typical Static Margins for Various Aircraft

Aircraft	Static Margin
Cessna 172	0.19
Learjet 35	0.13
Boeing 747	0.27
P-51 Mustang	0.05
F-16C	0.01
X-29	-0.33

Neutral Point

The **neutral point** (x_n) is the c.g. location at which the aircraft has **neutral static stability** ($C_{M_\alpha} = 0$). It is defined by:

$$\bar{x}_n = \bar{x}_{ac} + \frac{C_{L_{\alpha_t}}}{C_{L_\alpha}} \left(1 - \frac{\partial \varepsilon}{\partial \alpha}\right) V_H \tag{36}$$

Where $\partial\varepsilon/\partial\alpha$ is the downwash gradient at the tail.

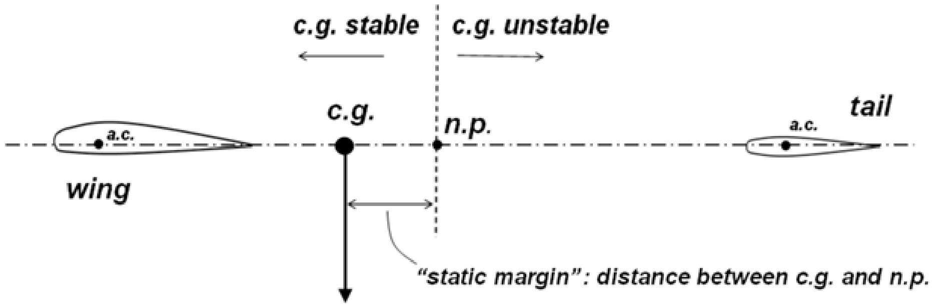

Figure 7: Static longitudinal stability: for a statically stable aircraft, the neutral point (n.p.) is aft of the center of gravity (c.g.). The static margin is the distance between them.

Static Margin

The **static margin** (SM) is a measure of longitudinal stability:

$$\boxed{SM = \bar{x}_n - \bar{x}_{cg} = -\frac{\partial C_M}{\partial C_L}} \tag{37}$$

- **Larger SM** $\rightarrow morestable, lessagile$
- **SM** $\approx$ **0** $\rightarrow neutrallystable$
- **SM <0** $\rightarrow staticallyunstable$

Scissor Plot (S_t/S)

The **scissor plot** is a design tool that plots the required tail-to-wing area ratio (S_t/S) as a function of the c.g. location ($\bar{x}_{cg}$). It establishes the forward and aft c.g. limits based on three criteria:

1. **Forward limit (stability):** ensures adequate static margin.

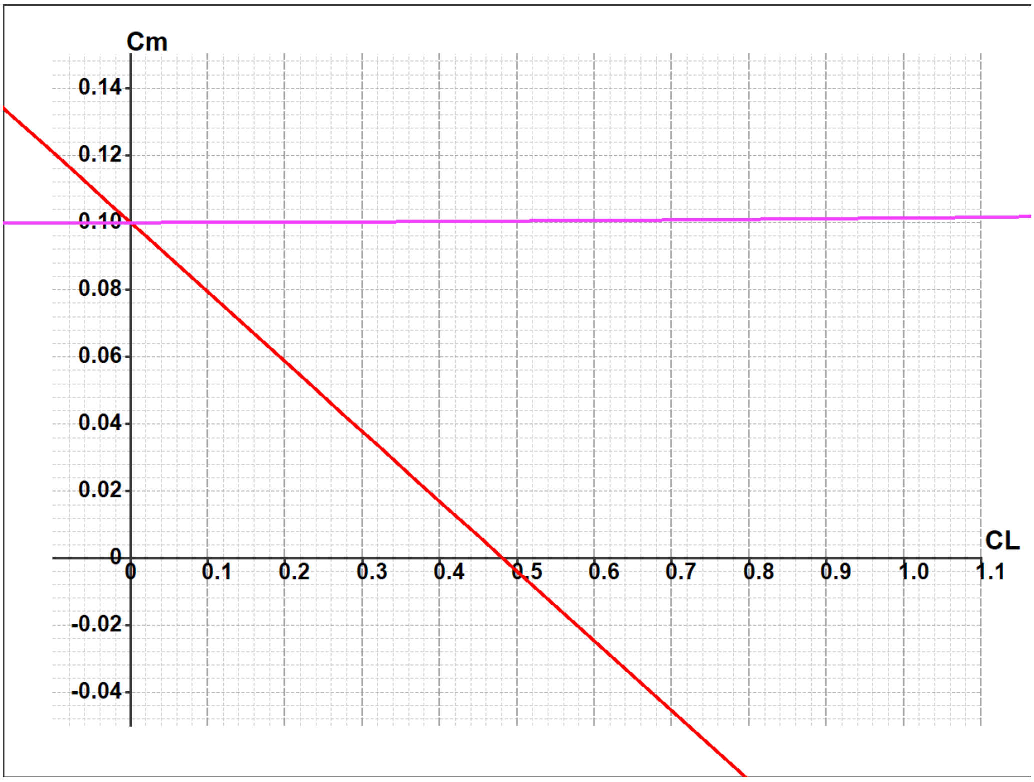

Figure 8: Example pitching moment (C_M) vs. lift coefficient (C_L) curve. The slope dC_M/dC_L is the negative of the static margin.

2. **Stall recovery:** ensures sufficient nose-down control authority at stall.
3. **Take-off rotation:** ensures sufficient nose-up moment to rotate during take-off.

The intersection of these constraints determines the minimum required tail size.

Trim Condition

An aircraft is **trimmed** when it does not rotate about the c.g. in flight. Two conditions must be satisfied simultaneously:

$$C_M = 0 \qquad \text{and} \qquad C_L = C_{L_{trim}} = \frac{2W}{\rho V_\infty^2 S_w} \tag{38}$$

Two unknowns must be solved: the angle of attack α_{trim} and the elevator deflection $\delta_{e_{trim}}$.

$$\alpha_{trim} = \frac{C_{M_0} C_{L_{\delta_e}} + C_{M_{\delta_e}}(C_{L_{trim}} - C_{L_0})}{C_{L_\alpha} C_{M_{\delta_e}} - C_{L_{\delta_e}} C_{M_\alpha}} \tag{39}$$

$$\delta_{e_{trim}} = -\frac{C_{M_0} C_{L_\alpha} + C_{M_\alpha}(C_{L_{trim}} - C_{L_0})}{C_{L_\alpha} C_{M_{\delta_e}} - C_{L_{\delta_e}} C_{M_\alpha}} \tag{40}$$

Where the elevator effectiveness derivatives are:

$$C_{L_{\delta_e}} = \frac{S_t}{S_w} C_{L_{\delta_{e_t}}}, \qquad C_{M_{\delta_e}} = C_{L_{\delta_{e_t}}} \frac{S_t}{S_w}(\bar{x}_{cg} - \bar{x}_{ac,w}) - C_{L_{\delta_{e_t}}} V_H \tag{41}$$

Trim Drag

Trim drag is the additional drag resulting from the deflection of the horizontal stabilizer and elevator to maintain the trim condition. Since the tail typically produces a downward force, the wing must produce extra lift to compensate, resulting in a slight increase in induced drag:

$$C_D = C_{D_0} + \frac{C_{L_w}^2}{\pi \cdot e_w \cdot AR_w} + \frac{C_{L_t}^2}{\pi \cdot e_t \cdot AR_t} \cdot \frac{S_t}{S_w} \tag{42}$$

Suggested Reading

- **Raymer:** Ch. 6
- **Gudmundsson:** Ch. 24 and 25
- **Drela:** Basic Aircraft Design Rules
- **Nelson:** Flight Stability and Automatic Control
- **Sadraey:** Ch. 5